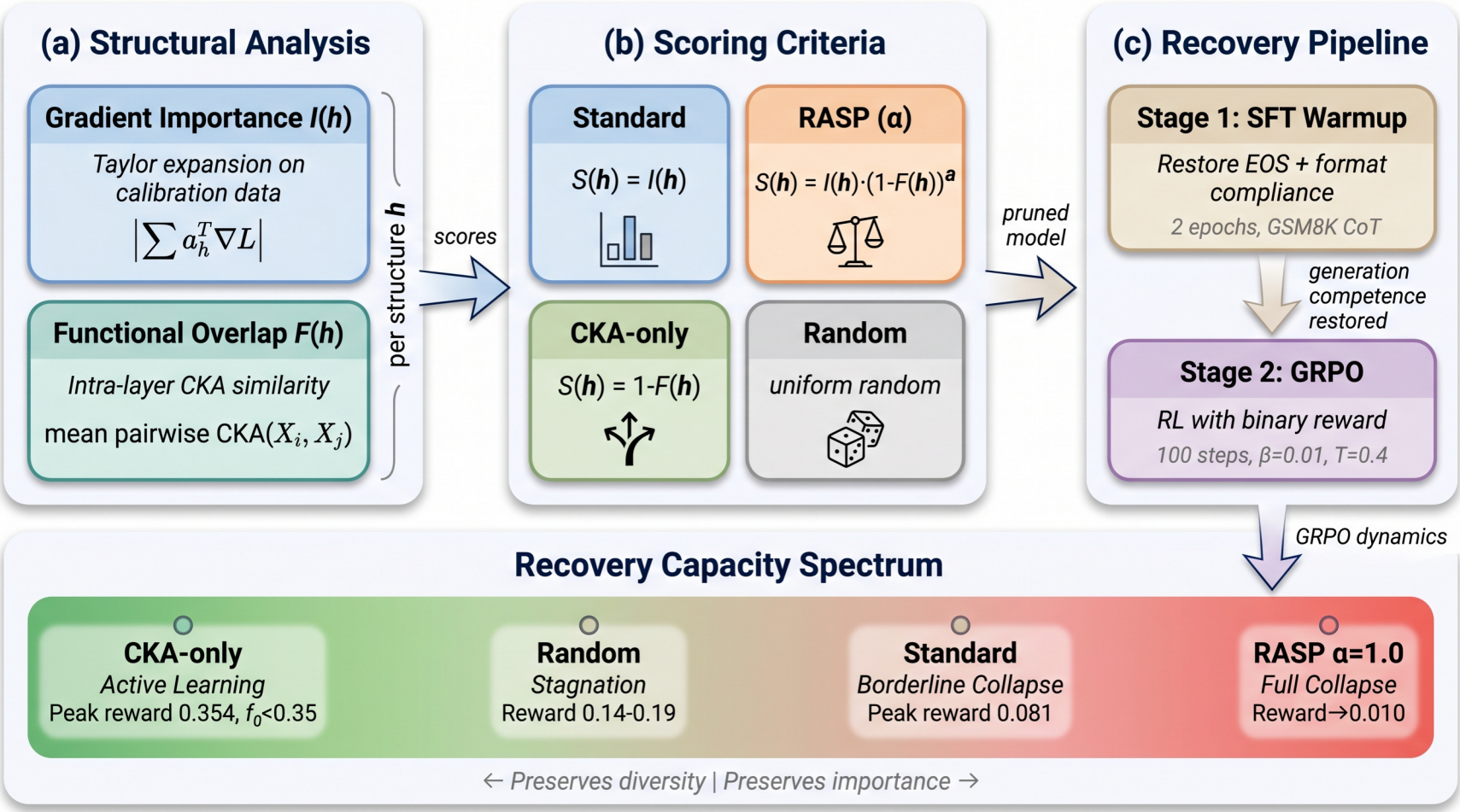

(a) Structural Analysis
Gradient Importance I(h)
Taylor expansion on calibration data
$\left|\sum a_h^T \nabla L\right|$
Functional Overlap F(h)
Intra-layer CKA similarity
mean pairwise CKA(X_i, X_j)
per structure h
scores
(b) Scoring Criteria
Standard
S(h) = I(h)
RASP (α)
S(h) = I(h)·(1-F(h))^a
CKA-only
S(h) = 1-F(h)
Random
uniform random
pruned model
(c) Recovery Pipeline
Stage 1: SFT Warmup
Restore EOS + format compliance
2 epochs, GSM8K CoT
generation competence restored
Stage 2: GRPO
RL with binary reward
100 steps, β=0.01, T=0.4
GRPO dynamics
Recovery Capacity Spectrum
CKA-only
Active Learning
Peak reward 0.354, f_0<0.35
Random
Stagnation
Reward 0.14-0.19
Standard
Borderline Collapse
Peak reward 0.081
RASP α=1.0
Full Collapse
Reward→0.010
← Preserves diversity | Preserves importance →